

Web Development

Introduction

This lecture is based on
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Why
are
YOU
in
this
class?

Course Objectives

- **To Understand**

- What is the Internet
- Is it the same, the WWW and the internet?
- How is the web architected?
- What is Internet Protocol?
- What is a domain?
- What are technologies surrounding the modern web?
- Data Standards on the Web
- Web Services Architecture
- What is the Full Stack Development

Background

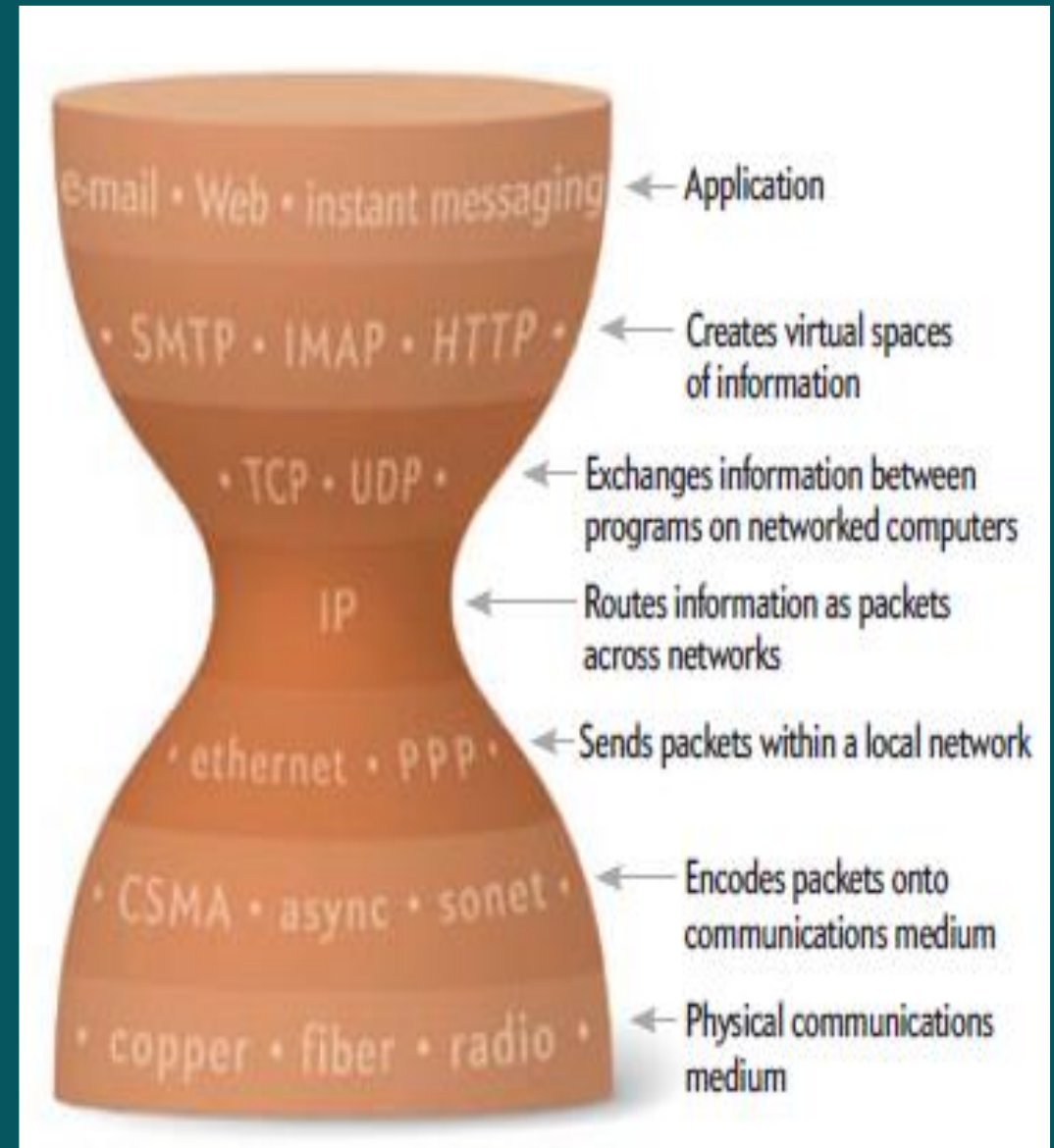
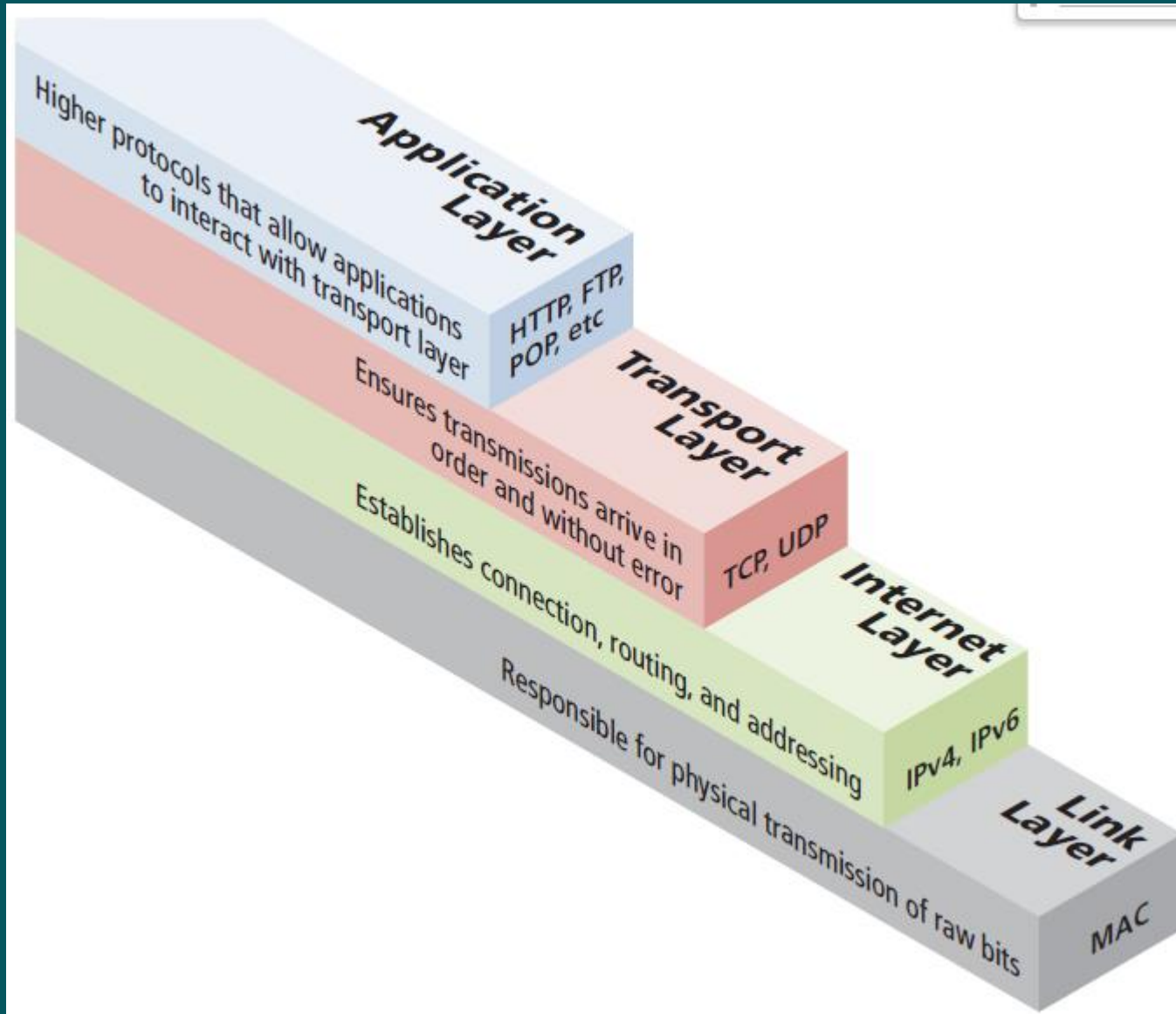
- What is the Internet
- Is the WWW is the internet?
- What is Internet Protocol?
- What is a domain Name?



The Stuff We Will Do

- Project Based
- Assignments : 20%
- 7th week Exam : 20% marks
- 12th week Exam : 20% marks (Project)
- • Final Exam : 40%

Network Stack

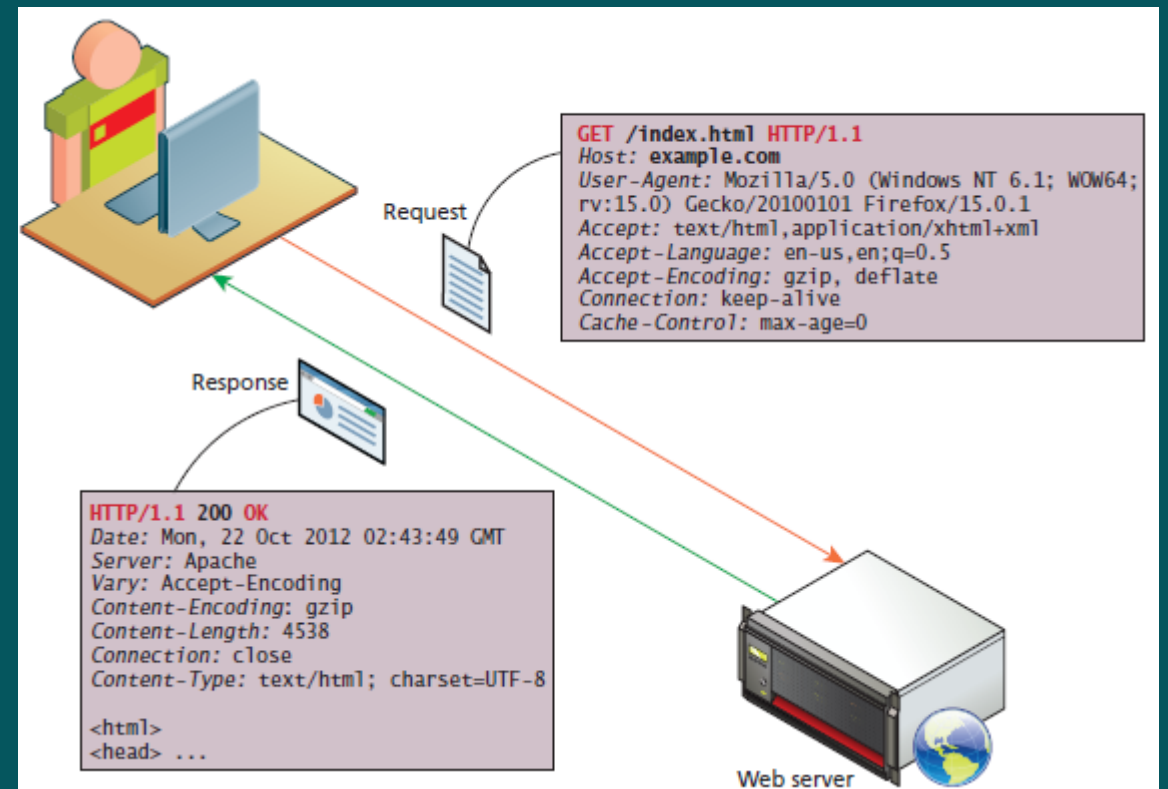


Internet

- Longest running computer networks in the world
- Pieces can be incremented individually
- Built on top of TCP/IP

HTTP – The Internet Protocol

- Protocol conveys **Data**
- Data can be interpreted as **Information**



HTML and CSS

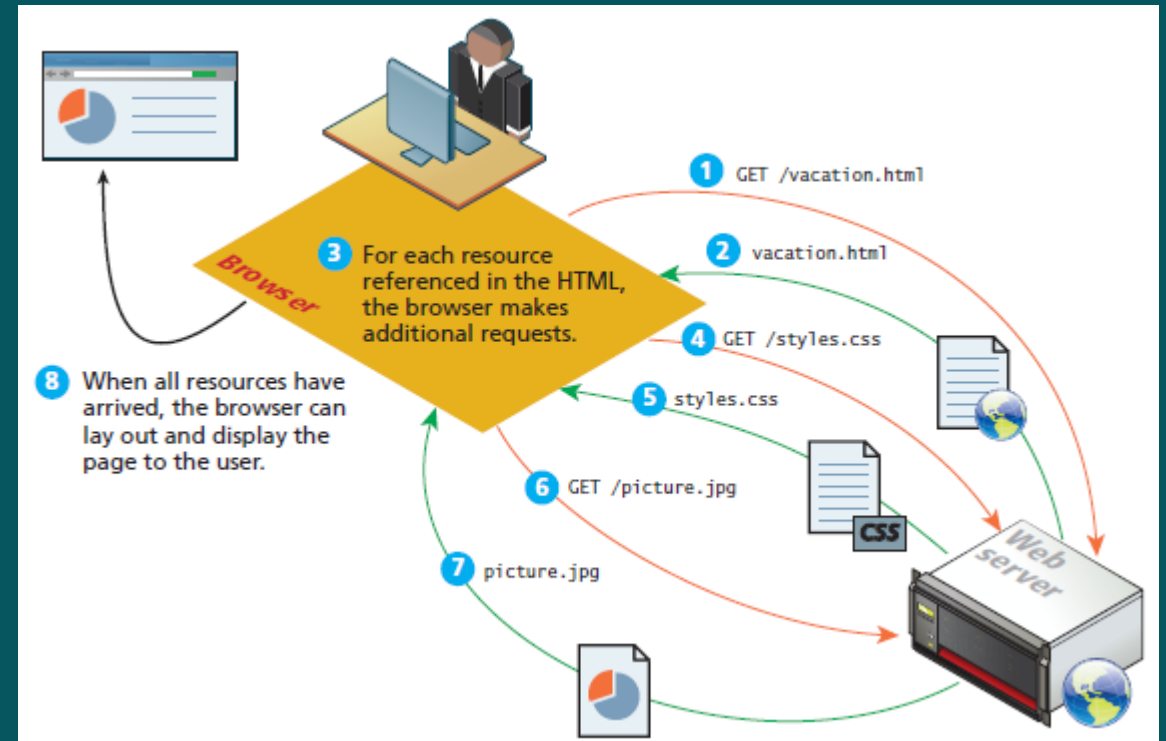
- HTML – Hyper Text Markup Language
 - **Hypertext** – Text with Links (as opposed to Linear)
 - **Markup** – Instructions included with the information
- CSS – Cascading Style Sheets
 - Style and modify the **look** of the HTML

Scripting

- JavaScript is the most popular front-end scripting language for the web
- General Programming Language – But for the **web** in mind
- Can “edit” HTML **live** on the web
 - Create new elements
 - Modify existing elements
 - Delete or Hide existing elements

Browsers

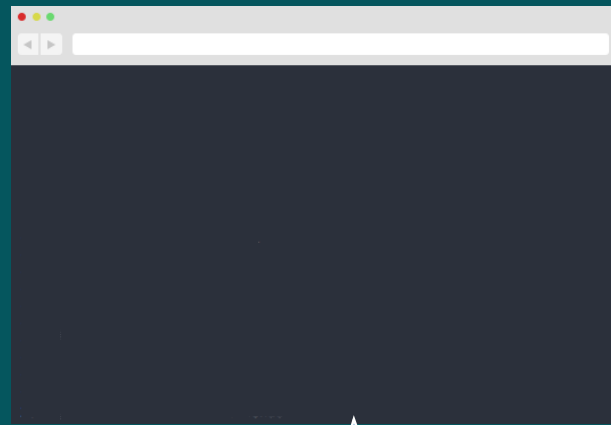
- Network Access
- Interpret the fetched data
 - Possibly fetching and interpreting dependencies like images
- Rendering HTML with CSS Rules
- Running scripts



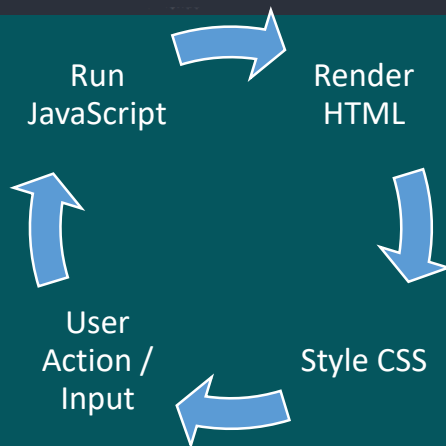
Browsers – How they operate

- Analyze the address bar to determine protocol and server
 - <http://www.google.com>
- Connect to server and download all dependencies
 - Dependencies – Images, CSS, JavaScript
- Analyze HTML and generate the **Document Object Model (DOM)**
- Apply CSS to DOM
- Display the DOM
- Start executing the script and manipulate the DOM as required

Typical Web Communication



HTML
CSS
JavaScript
Images



Question

Websites **vs** Web Apps

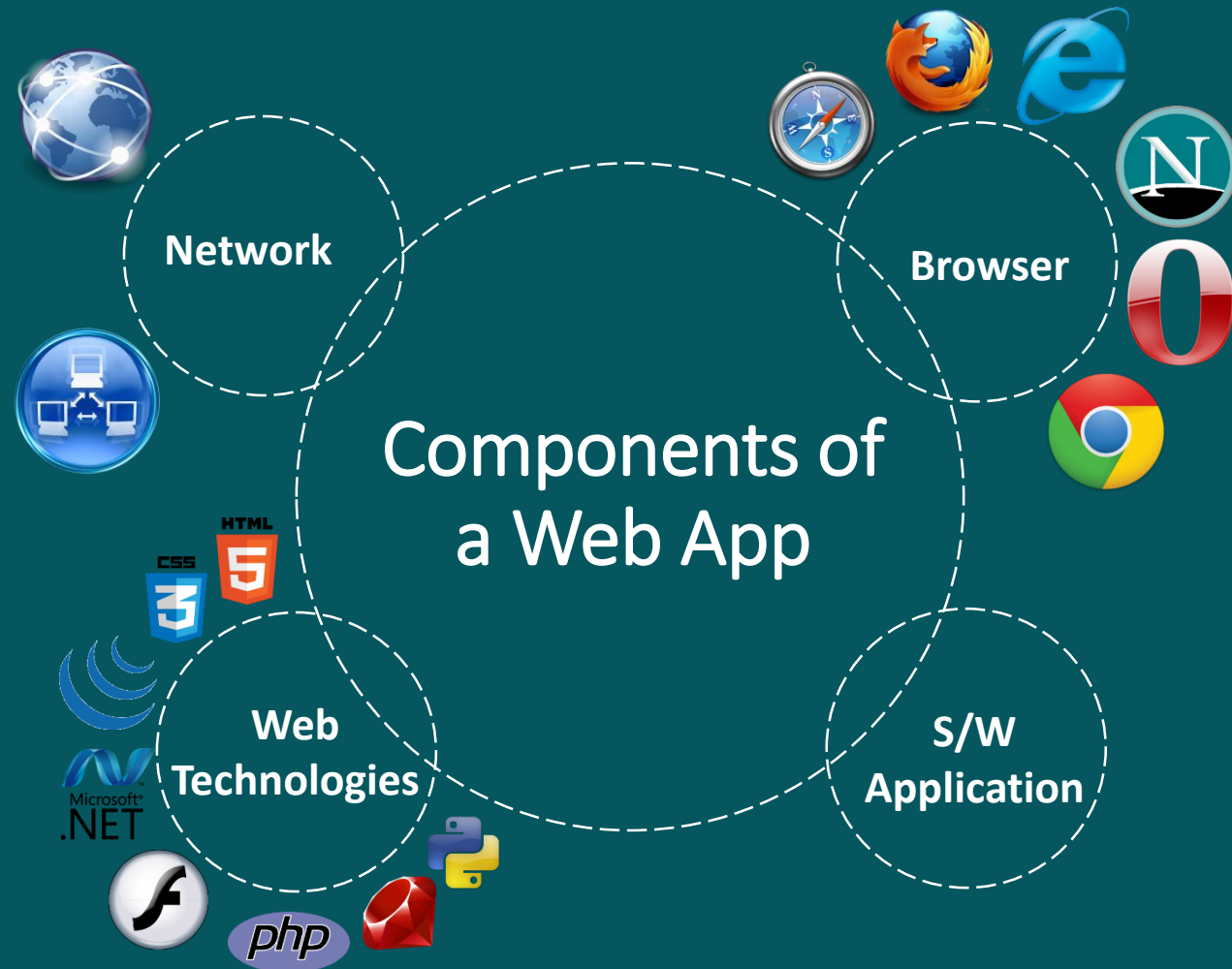


Websites vs. Web Applications

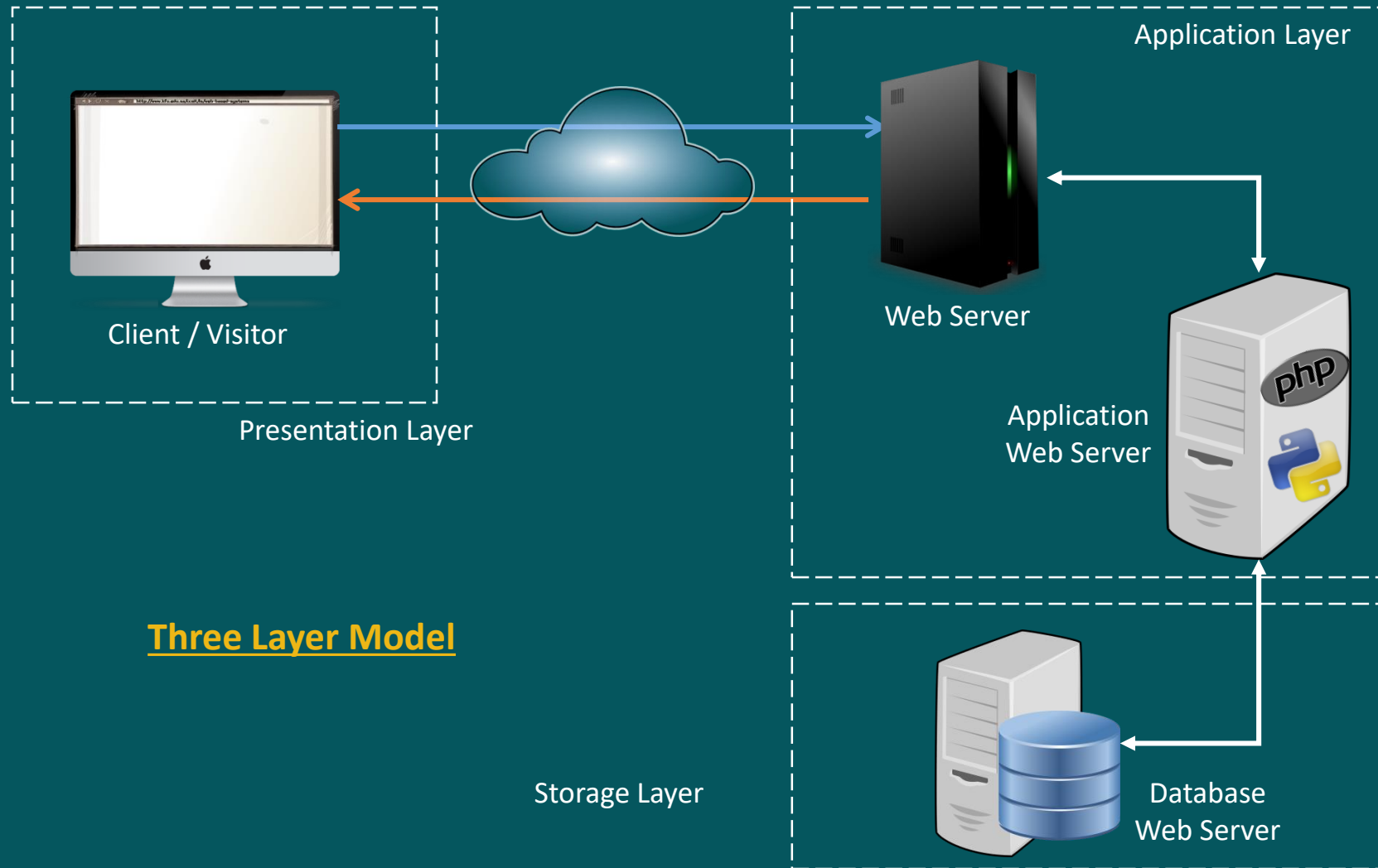
- Static
 - Function: Display content
 - No programming language or DB required
 - No or Limited User Interaction Required
 - Not (often) updated
- Dynamic
 - Function: Perform Tasks and Transactions
 - Web programming language & DB required
 - Rely on the user to interact
 - Ever-changing

Web Applications

- *A web application is an application utilizing web and [web] browser technologies to accomplish one or more tasks over a network, typically through a [web] browser*
- Also referred to as Web Apps or Web Widgets



How Web Applications Work



Types of Web Apps

Searching



Communication



Multimedia



Publishing



Social N/W



Shopping



Desktop Alt.

The +ve & the -ve

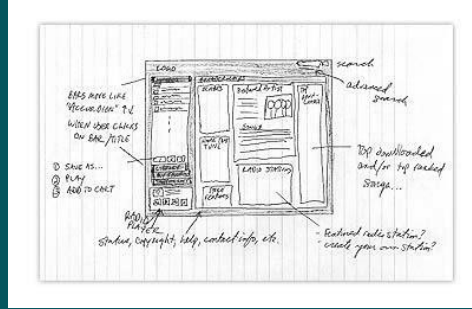
Benefits

- Platform-independent (Any Hardware & Any OS)
- No download, installation or update required
- Background objects such as databases are not on the user's computer

Drawbacks

- Security
- Must be available 24x7
- Testing done is not exhaustive
- Anything on the web stays on the web FOREVER

PLAN



MEET



THE PROCESS

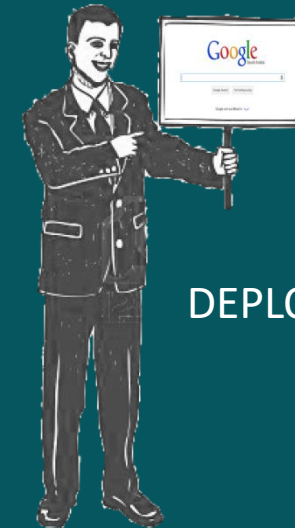
DESIGN



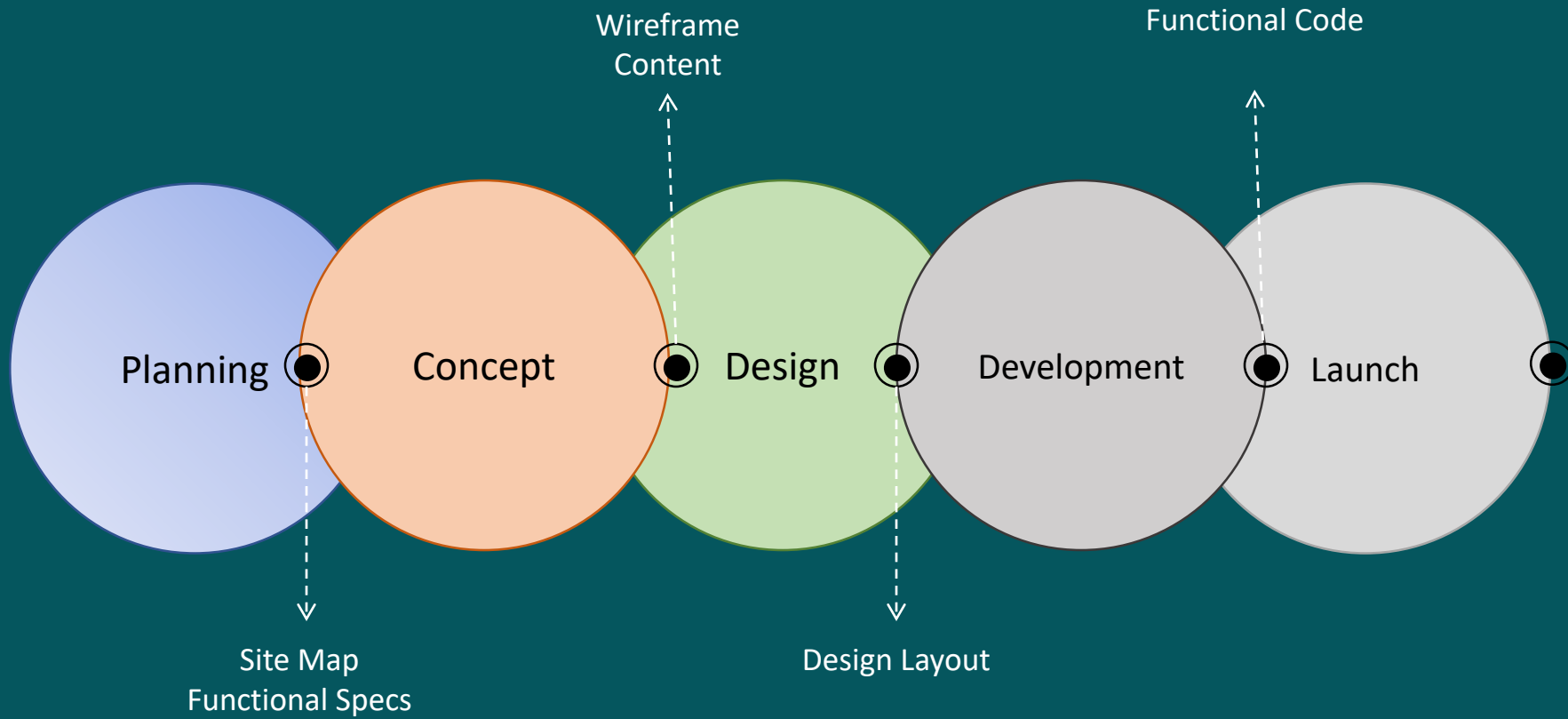
PROGRAM



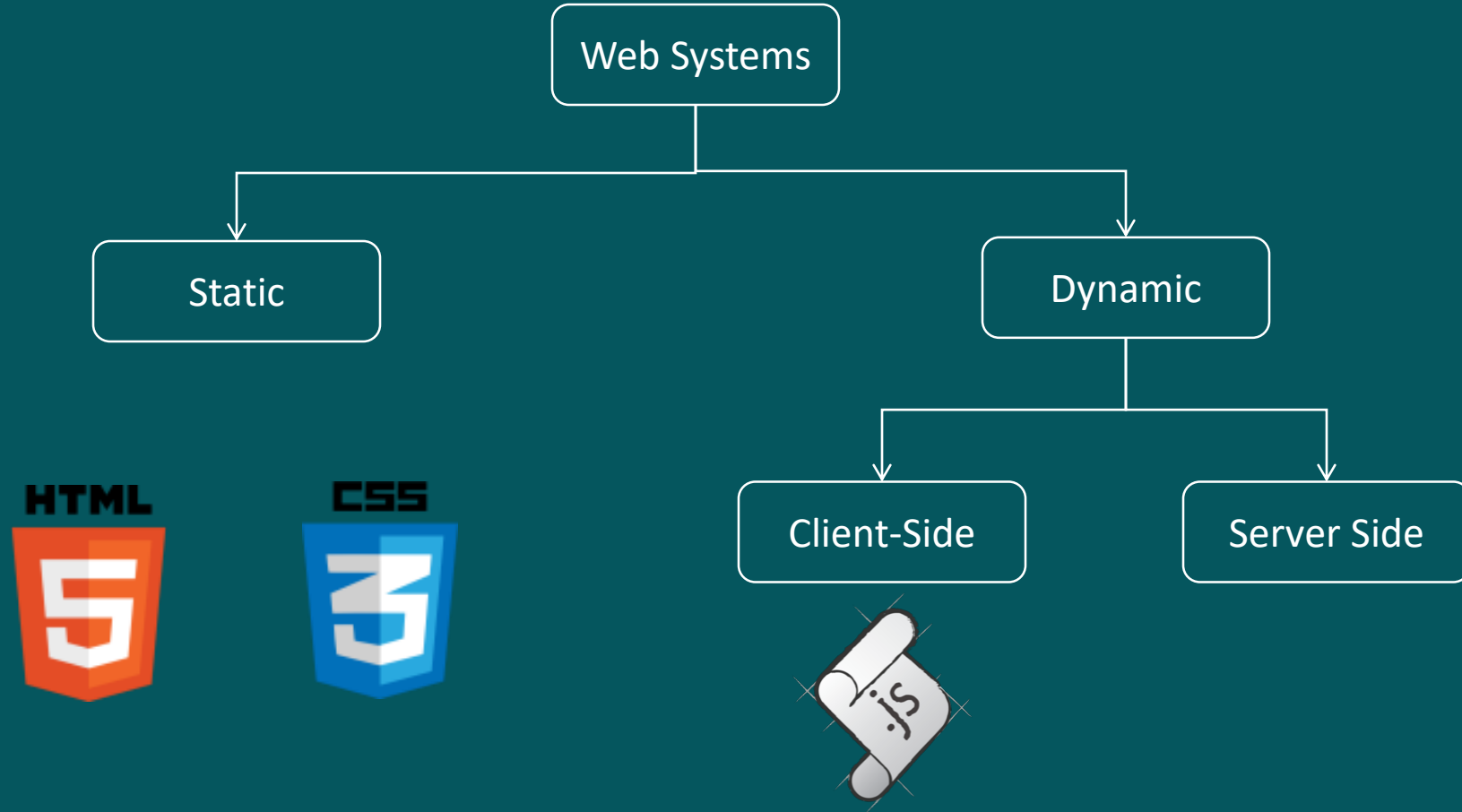
DEPLOY



The Process



Technology Framework



Server-Side Web Development Languages



General-purpose language designed for web development to produce dynamic web pages



Scribd.

Ruby is an OOP language designed to make programming fun and flexible



Python is a high level language designed to emphasize productivity and code readability



YouTube



ASP .Net is a web application framework that allows development of dynamic websites, web apps and web services



What is the Full Stack Development???

The process of developing both the frontend and backend of applications.

Web Developers

Front-End Developer / Designer

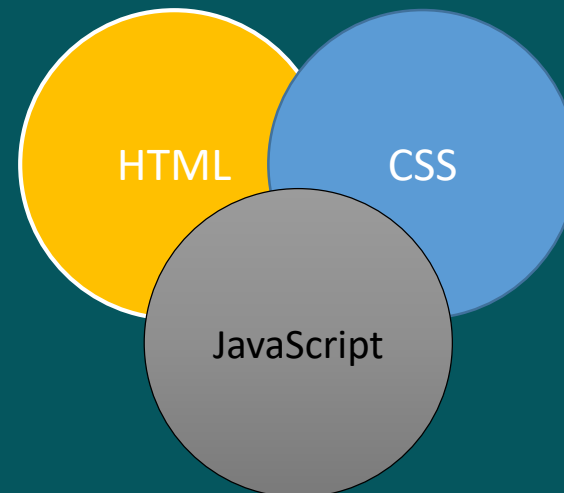
Back-End Developer / Web Programmer

Full Stack Developer



Front-End Developer

- Focuses on look and feel of a website
- Uses HTML, CSS, and JavaScript
- Is competent in color scheming, graphic design, and information flow
- Creates a great user experience
- Is right-brained: Strong intuition, creativity, & imagination



Back-End Developer

- Creates the inner workings of a website
- Is competent in programming languages (PHP, .NET, Python, C, Ruby)
- Is left-brained: logic, linear thinking, technical
- Hands-on coding experience is required

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Full Stack Developer

- Comfortable working with both back-end and front-end technologies.
- Work with
 - Databases
 - PHP/Python/.Net/Ruby/Java
 - HTML
 - CSS
 - JavaScript
 - *and everything in between*

What is a Full Stack?

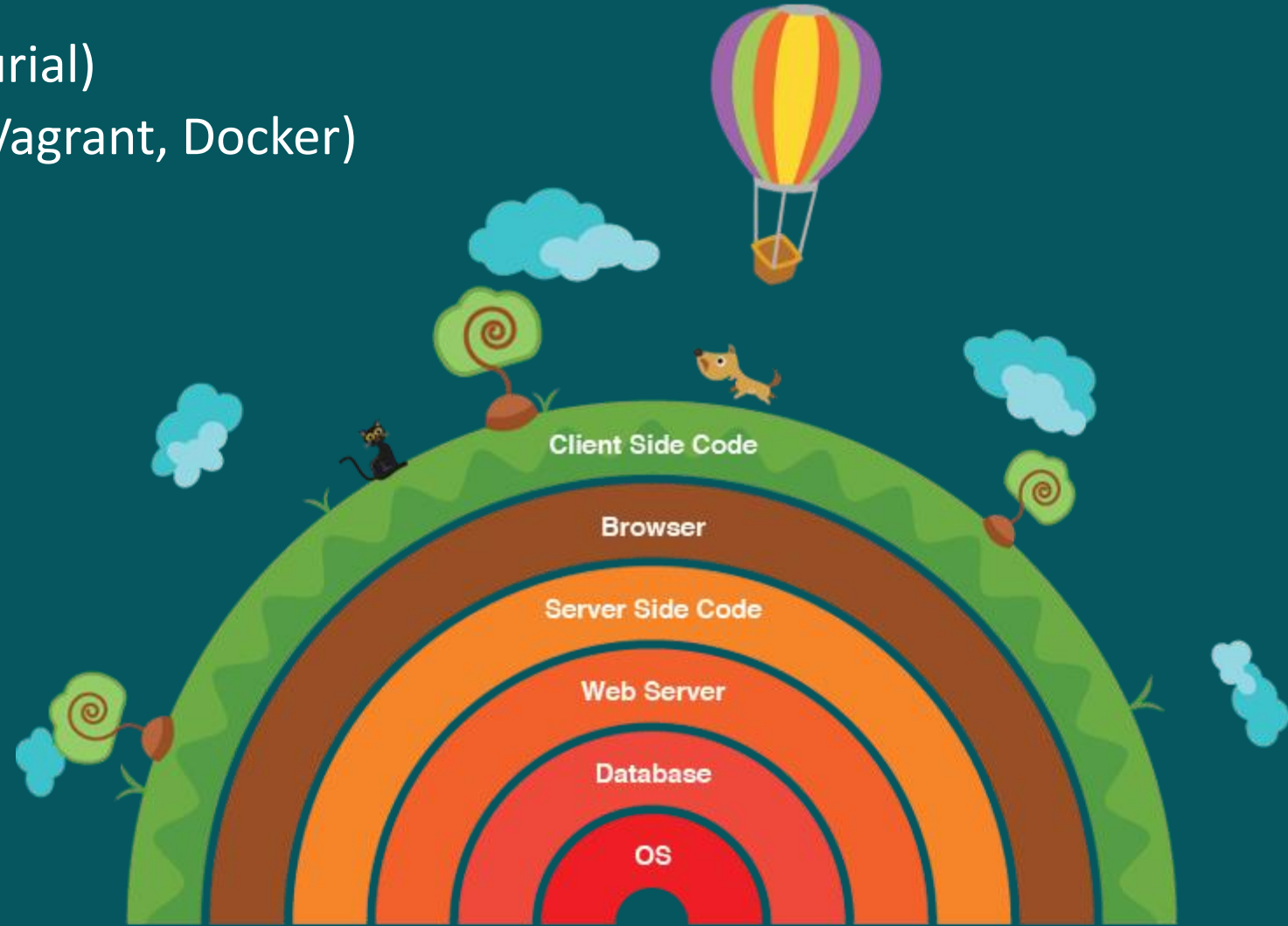
- **System Administration**

- Shell Scripting (Linux)
- Cloud Computing (Amazon, Rackspace)
- Search Engine Integration (ElasticSearch, Sphinx)
- Caching
- Monitoring



What is a Full Stack?

- **Web Development Tools**
 - Version Control (Git, Mercurial)
 - Virtualization (VirtualBox, Vagrant, Docker)



What is a Full Stack?

- **Back-end Technologies**

- Web Servers (Apache, IIS, Nginx)
- Programming Languages (PHP, NodeJS, Ruby)
- Databases (MySQL, PostgreSQL, MongoDB, Redis) – General (SQL, JSON, XML)



What is a Full Stack?

- **Front-end Technologies**
 - HTML / HTML5 Semantic Web
 - CSS / CSS3: LESS, SASS, Media Queries
 - JavaScript: JQuery, AngularJS, ...
 - Browser Compatibility
 - Responsive Design
 - AJAX, JSON, XML, WebSocket



Assignement

RESTful API **vs** Microservices

